

Data Scientist Assessment

Test Duration: 3 Days

kumparan's Dataset & Model Interface

You are given a dataset of articles in file `data.csv` and model interface file `model.py`.
The `data.csv` file contains 3 columns:

- `article_id`: unique id of the article
- `article_topic`: main topic of the article
- `article_content`: content of the article

You will implement the task explained below inside file `model.py`.

The dataset and the Kumparan's model interface can be downloaded via:

Task Overview

From the dataset, you should be able to create a topic extractor model that basically extracts a topic given the content of the article. We expect the topic extractor to behave as the dataset given. So given the `article_content` it will output the `article_topic`.

To solve the task, please do the following:

1. Implement the task in Kumparan's model interface `model.py` (available in `ds_assesment_v2.zip`).
 1. You can use any external library such as `pytorch`, `tensorflow`, `scikit-learn` etc.
 2. Implement your training procedure inside `.train()` method (see `model.py`).
 3. Modify the `.predict` method that takes a single row `article_content` (a string) as the input and outputs the topic name of the article based on your model prediction result. We should be able to use your model as: `article_topic = model.predict(article_content)`
2. The implementation (training step) should run under 30 minutes and must run under 1 hour (*Using GPU for this task is not allowed*).
 1. Running time longer than 30 minutes will have a minus score.
3. You are not allowed to use a pre-trained model.
4. You must write the documentation of your implementation.
 1. The documentation must include the step-by-step on how to set up and run your implementation.
 2. The documentation must be in the plain text, markdown, or PDF format.
 3. The documentation should include a brief explanation on the method and why you use it.
 4. The documentation should include the metric you use to measure / validate the correctness of the method.

5. If you are given more time, what kind of improvement do you want to implement? Write your answer on the documentation as well.

We will run your trained model on our private validation dataset for scoring.

Submission

It should work in **Python 3.11**.

Generate the model.pickle by running your modified model.py file:

```
python model.py
```

It should run your training procedure and save your trained model to model.pickle file.

To submit your work, zip the following required files:

- Documentation files (.txt, .pdf or .md)
- Your modified model.py files
- Your trained model (model.pickle)

The zip file should be named using the following format:

ds_<your_name_with_underline>.zip

and submit it.

Scoring Method

Here are the priority of the scoring method for the task:

1. The accuracy of the model on the validation dataset.
2. The flow and logic for the method (in the documentation).
3. How clear is the code and the explanation?
4. Future improvement.

Also in this case, please let us know the reason you can't do the task (whether the time given is not enough, you are sick, or something else).

Note on AI Usage

We recognize that AI tools (such as ChatGPT, Gemini, Claude, Copilot, etc.) are increasingly used in technical work and exploration. Therefore, **we allow the use of AI to help complete this assessment**, with the following important conditions:

- **Transparency is required:** Please clearly indicate which parts of your work were assisted by AI. For example, whether AI was used for idea exploration, initial code generation, documentation writing, or other tasks.

- **You are fully responsible:** You remain accountable for the quality, accuracy, and understanding of the solution you submit. We will evaluate not just the final result, but also your grasp of the approach you used.
- **We'll know:** Our team has also tested this assessment using various AI tools. So we'll be able to tell if you simply copied AI output without meaningful thought or adaptation.

The main goal of this assessment is to understand your thinking and problem-solving approach. AI can assist, but **it cannot replace your own critical thinking and understanding.**

Timeline & Expectations

You have **3 days** to complete this assessment. While AI tools might help you finish faster, our priority is **quality over speed**. Use the time wisely to refine your approach, validate your results, and ensure your submission reflects your own thinking and understanding.

The goal of this assessment is to understand how you think and solve problems, not just how fast you can generate output.